

## *Calcified neurocysticercosis among patients with primary headache.*

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BACKGROUND: Anecdotal reports and a single case-control epidemiological survey have suggested an association between the helminthic disease neurocysticercosis and primary headache. The present study was undertaken to determine whether neurocysticercosis is more common among patients with primary headaches than in other neurological disorders. METHODS: We determined the prevalence of neurocysticercosis in a cohort of patients with primary headache who were seen at our institution over a 20-year period. We used as controls all people from the same cohort with four major different categories of neurological disorders, including cerebrovascular disease, degenerative disorders of the CNS, head trauma, and primary brain tumors. We evaluated differences in the prevalence of neurocysticercosis between patients and controls. RESULTS: Forty-eight of 1017 patients with primary headache and 31 of 1687 controls had neurocysticercosis (4.7% vs 1.8%,  $p < 0.0001$ ). Calcified parenchymal brain cysticerci were more frequent among patients with primary headache than in those with cerebrovascular disease (4.7% vs 1%,  $p < 0.001$ ), degenerative disorders of the CNS (4.7% vs 2.4%,  $p < 0.05$ ), and head trauma (4.7% vs 2.3%,  $p < 0.05$ ). There were no significant differences, however, for the subset of controls with primary brain tumors (4.7% vs 3.5%), a condition that has also been associated with neurocysticercosis. CONCLUSIONS: There is a relationship between calcified neurocysticercosis and primary headache disorders. It is possible that periodic remodeling of cysticercotic calcifications, with liberation of antigens to the brain parenchyma, contributes to the occurrence of headache in these patients.